R.F. Source F.E.T. Replacement Procedure

CLS Test Procedure – 1.7.32.2 Rev. 1

Date: 2008-09-03

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REVISION HISTORY

Revision	*Date*	*Description*	*Author*
A	29-Mar-07	Original Draft	D.J. LeClair
B	17-Apr-07	Added Tables and Drawing	D.J. LeClair
0	20-June-07	Released for Use (Tested and Verified)	D.J. LeClair
1	2008-09-03	Added additional FET Test Clarified some text	H. Buchmann

TABLE OF CONTENTS

1.0 PURPOSE

This document outlines the procedure for replacing the F.E.Ts in the R.F. Source.

1.1 BACKGROUND

Regular preventive maintenance can prevent catastrophic failure of components of the R.F. Source.

1.2 QUALIFICATIONS

Electronic Technician, Technologist or Engineer.

1.3 TEST EQUIPMENT

50 ohm termination

Digital V.O.M.

5/16 inch wrench

2.0 PROCEDURE FOR R.F. SOURCE F.E.T. REPLACEMENT

Lock out and tag the AC for the RF Source at panel DPNL 1021-01. Verify that the AC is off. The RF Source is fed via a plug which could be plugged into the wrong outlet. It's best to remove the plug.

To remove the F.E.Ts from the R.F. Source, first turn off the H.V. Power Supply. Open the rear door and make sure that the shorting bar has properly shorted out the H.V. Power Supply. Disconnect the trigger cable that runs through the toroids that are attached to the F.E.Ts. Pull out the F.E.T. circuit boards.

To test the F.E.Ts, place the circuit board on its side. Turn on the V.O.M. and set it to the diode check position. Place one probe of the V.O.M. on the copper pad on each side of the board (spring loaded contacts). Reverse the probes. One direction should read about 0.5 on the meter. The other direction should be infinite. Lay the board down with the traces facing upwards. Check the driver F.E.Ts (IRFD110) on the board by placing the probes on the source and drain of each F.E.T., in both directions. The readings should be similar to the first set of readings, i.e. 0.5 in one direction and infinite in the other.

Measure the resistance across the 1 MΩ resistor. In either direction the resistance should read more than 800 KΩ. A reading of less than 800 KΩ is indicative of a leaky or faulty power F.E.T. (APT1002RBN).

Replace any questionable or bad F.E.T and return the circuit boards to the chassis. There are arrows on the F.E.T. toroids indicating the trigger path. The arrow goes from left to right on the top row and right to left on the bottom row. Ensure that the F.E.Ts are in the proper configuration. Reconnect the trigger cable and close the back door. The trigger cable is strung as follows: The yellow end of the cable goes to the centre jack of the patch panel on the left side of the rack. The cable goes through the toroids on the upper row of F.E.T.s, from left to right, through the toroids on the bottom row of F.E.Ts, from right to left, to the patch panel jack closest to the back door. See the drawing at the end of this procedure.

Restore AC to the R.F. Source by removing the lock and tag. Turn on the H.V Power Supply. Wait for the time delay.

The klystron in the R.F. Drive is rated for 8 kV C.W. operation. For pulse operation, we operate the klystron at 15 kV. After replacing the F.E.Ts in the R.F. Drive, the klystron should be gradually returned to full power operation using the following procedure.

Disable the drive to the Modulators (EPICS RunLinac Screen).

On the R.F. Drive:

Turn off the High Voltage Power Supply.

Terminate the output of the R.F. Source amplifier with a 50 ohm termination.

Turn the Kilovolts control on the H.V. Power Supply fully CCW.

The rest of this procedure requires that the LINAC Zones 1-5 are locked up.

Enable the R.F. (EPICS RunLinac Screen).

Turn on the High Voltage Power Supply

Turn the Kilovolts control on the H.V. Power Supply CW until the meter reads 6 kV. Leave it for 5 minutes.

Increase the Kilovolts to 10. Leave it for 5 minutes.

Increase the High Voltage 500 volts at a time until 15 kV has been reached. Wait 5 minutes at each level.

Turn off the High Voltage Power Supply.

Reconnect the output of the R.F Source amplifier.

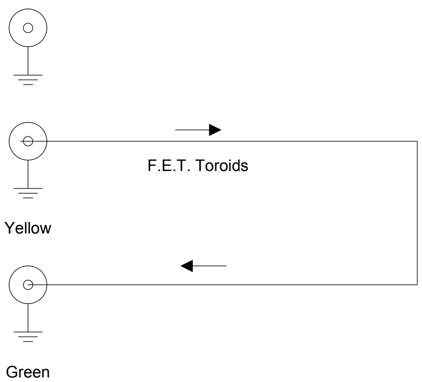
Turn the R.F. attenuator several turns CW.

Turn on the High Voltage power supply.

Gradually peak the R.F. Drive over a period of approximately ten minutes by turning the R.F. Attenuator CCW.

Malfunctioning that could damage the F.E.Ts will be noted by jitter in the High Voltage pulse and by fluctuations in the High Voltage meter on the power supply. The R.F. Drive should be constantly monitored during the run-up procedure. Any malfunction should be investigated. If no cause is found, the procedure should be resumed at lower drive levels and with a more gradual run-up.

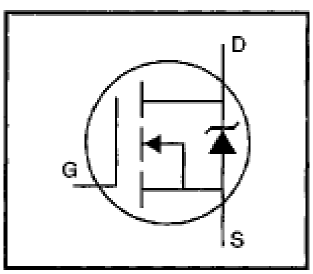
Trigger Path.

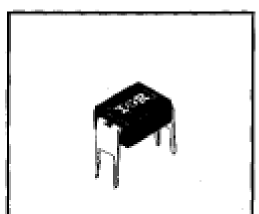


IRFD110

Absolute Maximum Ratings

	Parameter	Max.	Units
I_D @ T_C = 25°C	Continuous Drain Current, V_{GS} @ 10 V	1.0	A
I_D @ T_C = 100°C	Continuous Drain Current, V_{GS} @ 10 V	0.71	
I_{DM}	Pulsed Drain Current ①	8.0	
P_D @ T_C = 25°C	Power Dissipation	1.3	W
	Linear Derating Factor	0.0083	W/°C
V_{GS}	Gate-to-Source Voltage	±20	V
E_{AS}	Single Pulse Avalanche Energy ②	140	mJ
I_{AR}	Avalanche Current ①	1.0	A
E_{AR}	Repetitive Avalanche Energy ①	0.13	mJ
dv/dt	Peak Diode Recovery dv/dt ③	5.5	V/ns
T_J T_{STG}	Operating Junction and Storage Temperature Range	-55 to +175	°C
	Soldering Temperature, for 10 seconds	300 (1.6mm from case)	





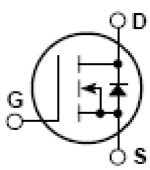
HD-1

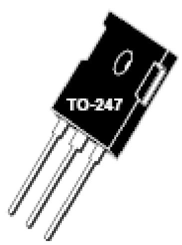
APT1002RBN

N-CHANNEL ENHANCEMENT MODE HIGH VOLTAGE POWER MOSFETS

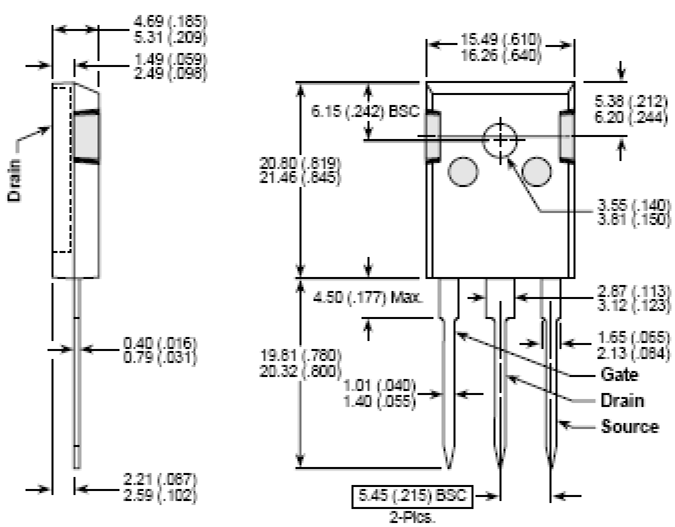
MAXIMUM RATINGS All Ratings: T_C = 25°C unless otherwise specified.

Symbol	Parameter	APT 1002RBN	APT 1002R4BN	UNIT
V_{DSS}	Drain-Source Voltage	1000	1000	Volts
I_D	Continuous Drain Current @ T_C = 25°C	7.0	6.5	Amps
I_{DM}	Pulsed Drain Current ①	28	26	
V_{GS}	Gate-Source Voltage	±30		Volts
P_D	Total Power Dissipation @ T_C = 25°C	240		Watts
	Linear Derating Factor	1.96		W/°C
T_J, T_{STG}	Operating and Storage Junction Temperature Range	-55 to 150		°C
T_L	Lead Temperature: 0.063" from Case for 10 Sec.	300		





TO-247AD Package Outline



Dimensions in Millimeters and (Inches)